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The modified-gravity arm: a Bekenstein–Milgrom field theory for $a_0 = \kappa c \sqrt{G\rho_\Lambda}$

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The theory in one box

$$S = - \int d^3x \left[\frac{a_0^2}{8\pi G} \mathcal{F}(X) + \rho \phi \right], \quad X = \frac{|\nabla \phi|^2}{a_0^2}, \quad \nabla \cdot [\mu(x) \nabla \phi] = 4\pi G \rho$$

with $\mu = \mathcal{F}'(X)$, $x = |\nabla \phi|/a_0$, and the acceleration scale **not** a free constant but

$$a_0 = \kappa c \sqrt{G\rho_\Lambda} = \frac{cH_\Lambda}{Z} = 9.36 \times 10^{-11} \text{ m s}^{-2}, \quad \kappa = \tfrac{1}{2}, \quad Z = 2\sqrt{8\pi/3}$$

The free function is not postulated. For the framework’s in-force kernel $\nu = 1/(1 - e^{-\sqrt{y}})$ it is **derived in closed parametric form**, and its derivative is exact at every acceleration:

$$\mathcal{F}'(X) = \mu = 1 - e^{-\sqrt{y}}, \quad y = x\mu = g_{\text{bar}}/a_0$$

with limits $\mathcal{F} \rightarrow \frac{2}{3}X^{3/2}$ (deep) and $\mathcal{F} \rightarrow X - C_0 + 4e^{-s}(s^3 + 3s^2 + 6s + 6)$, $s = X^{1/4}$ (Newtonian), where C_0 is a pure constant in the Lagrangian and therefore **unobservable** — only $\mathcal{F}'$ enters the field equation.

Seven properties are proved rather than assumed (§4): convexity of $\mathcal{F}$ — from which existence, uniqueness, Newton’s third law, standard centre-of-mass motion and the exact baryonic Tully–Fisher relation all follow — plus ellipticity, ghost-freedom, subluminality, positive phantom density, and a Newtonian residual that is exponentially small *in the action*.

And it lenses, because the modification is in the field equation. That is why this is the arm.

The honest caption. $\kappa = \frac{1}{2}$ is **fitted**; nothing here derives it. Bekenstein–Milgrom is not new — it is theirs, from 1984. What is offered is the *coefficient*, the kernel forced by solar-system ranging, and the closed-form free function with its health proved for that kernel. The theory is non-relativistic as written (§7). And the cluster shortfall is untouched by the switch (§5.3).

0. Why this paper exists

This framework has spent most of its life as a **modified-inertia** theory: the metric Newtonian, the particle’s inertia acceleration-dependent. That reading is now excluded, and the exclusion is worth stating plainly because it is what motivates everything below.

In modified inertia the metric is unmodified and sourced by baryons — that is not an incidental feature but a requirement, since the whole content is a modified *response* to a Newtonian field. And photons have no rest mass to modify. So modified inertia predicts $M_{\text{lens}} = M_{\text{bar}}$, hence $M_{\text{dyn}}/M_{\text{lens}} = 1/f_{\text{bar}} \approx 6.4$ in clusters, against an observed 1.0–1.3: **excluded at $\sim 21\sigma$** , and still at 4.2σ if the systematic on that ratio is inflated fivefold [1].

Three rescue routes were computed and all three closed. The enhancement cannot sit in *both* the metric and the inertia — doing both gives $a = \nu^2 g_{\text{bar}}$, wrong by one factor of ν . The worldline action does have Randers–Finsler form when μ depends on position, which would give photons a geometry to follow — but **the resulting null cone degenerates at $\mu = \frac{1}{2}$ and the signature turns Euclidean below it**, unavoidably, since every interpolation reaching the deep-MOND limit must cross $\mu = \frac{1}{2}$; on this framework’s kernel that happens at $g_{\text{bar}} = 0.48a_0$, the outer disc of a spiral galaxy [2]. And Bekenstein–Milgrom’s own point-particle limit is *memoryless*, so the modified-inertia action is not that limit either [3].

Modified inertia and modified gravity are algebraically identical for massive test particles. That is precisely why the framework could remain ambiguous about which it was for years. Lensing is the observation that separates them, and it selects the metric.

What survives the switch is the part that mattered: $a_0 = \kappa c \sqrt{G\rho_\Lambda}$ **is a statement about the coefficient and is arm-independent.**

1. The coefficient claim

$a_0 \sim cH_0$ has been noticed for forty years. Writing it with the *asymptotic* de Sitter rate $H_\Lambda = c\sqrt{\Lambda/3}$ instead of the present one makes it a constant for a structural reason rather than a coincidence of epoch, and turns it into a relation between the **dark-energy scale** and the **dark-matter-phenomenology scale**:

$$a_0 = \kappa c \sqrt{G\rho_\Lambda}, \quad \kappa = \frac{1}{2} \Rightarrow a_0 = 9.3619 \times 10^{-11} \text{ m s}^{-2}$$

Λ CDM has no reason to predict such a relation; this framework has no freedom to avoid it. That asymmetry is the framework’s principal asset, and it is **not** evidence that the framework is correct.

The mechanism behind the numerical coincidence is Milgrom’s: an observer with proper acceleration a in de Sitter space sees a temperature $\propto \sqrt{a^2 + a_\Lambda^2}$ [4,5], so the bath has a floor, and the excess $\Delta T = T(a) - T(0)$ has both MOND limits. **The interpolating function that follows is Eq. (9) of Milgrom (1999), not a variant of it.** That paper fixes $a_0 = 2cH_\Lambda$; $\kappa = \frac{1}{2}$ corresponds to $cH_\Lambda/5.79$. The factor of eleven between them is this framework’s distinctive quantitative content, and it is fitted, not derived.

2. Why the kernel is exponential

MOND’s interpolating function is usually treated as free within limits. It is not: **solar-system ranging constrains its approach to Newtonian behaviour**, and two natural power-law families fail.

- The $1/g$ -tailed kernel misses the Earth/Mars bound by a factor ~ 1279 .
- The next one misses the Mars ranging budget by $8.5\text{--}12.4\times$, and for a subtle reason: its slowly-decaying tail binds not at a planet but at **the Sun**, whose acceleration in Jupiter’s field is itself only $\sim 2233 a_0$.

The surviving choice is an **exponential** approach,

$$\nu(y) = \frac{1}{1 - e^{-\sqrt{y}}}, \quad \mu = 1 - e^{-\sqrt{y}}$$

whose Newtonian residual dies as $e^{-\sqrt{y}}$ — utterly negligible at ephemeris accelerations ($\nu-1 < 10^{-40}$ at $y = 10^4$). This was arrived at against the author’s preference and it retired two previously published kernel choices [6].

3. The field theory, and the free function

Bekenstein–Milgrom [7] replaces the Poisson equation with $\nabla \cdot [\mu(|\nabla\phi|/a_0)\nabla\phi] = 4\pi G\rho$, derivable from the action in the box. The theory’s content is the free function $\mathcal{F}$, and for this kernel it is obtained rather than assumed.

In spherical symmetry the field equation integrates exactly to $\mu(x)x = y$, and $g_{\text{obs}} = \nu(y)g_{\text{bar}}$ means $x = \nu(y)y$. Substituting $u = \sqrt{y}$ gives the whole theory in one clean parametric pair:

$$\mu = 1 - e^{-u}, \quad x = \frac{u^2}{\mu}, \quad u \in (0, \infty)$$

from which $\mathcal{F}$, its convexity, the ellipticity eigenvalues, the scalar sound speed and the phantom density all follow analytically. **The map is a bijection and the inversion round-trips**, so the parametrisation is not a reparametrisation trick.

The two asymptotic forms are exact and quotable:

$$\mathcal{F}(X) \rightarrow \frac{2}{3}X^{3/2} \quad (\text{deep MOND}), \quad \mathcal{F}(X) \rightarrow X - C_0 + 4e^{-s}(s^3 + 3s^2 + 6s + 6), \quad s = X^{1/4} \quad (\text{Newtonian})$$

A trap worth flagging, because the accompanying script caught itself falling into it. $X = x^2$, so $\sqrt{X} = x$, and $x = y/\mu \sim u^2$ while $u = \sqrt{y}$. The Newtonian residual is therefore exponential in $X^{1/4} = \sqrt{x} = \sqrt{y}$, **not** in $\sqrt{X}$. The clean physical statement is that $\mathcal{F}'$ approaches unity like $e^{-\sqrt{g_{\text{bar}}/a_0}}$ — Route A’s kernel restated variationally.

And that is the field-theoretic point of the whole construction: the ephemeris relief is a property of the Lagrangian, not of an algebraic force law bolted on afterwards.

4. What is proved

Seven properties, established rather than assumed [8]:

convexity of $\mathcal{F}$	and from it: existence, uniqueness, Newton’s third law, standard centre-of-mass motion, exact BTFR
ellipticity	both eigenvalues of the field operator computed
ghost-freedom	
subluminality	the scalar sound speed computed
positive phantom density	
exact BTFR	with the coefficient obtained <i>from the action</i>
exponentially small Newtonian residual	in the action, evaluated at the Sun

Every standard theorem of the Bekenstein–Milgrom class follows from the first line. That is the structural reason this arm is on firmer ground than the one it replaces: the health is a *consequence* of a single proved property rather than a list of separately-checked conditions.

5. Observational standing

5.1 Galaxy scales: works, and non-diagnostic of the coefficient

The radial acceleration relation is reproduced with 0.108 dex scatter at $\Upsilon_{3.6} = 0.70$, *better* than regularised MOND’s 0.122 on the same data. But **the penalty for moving a_0 by 20% is under half a per cent in that scatter**. The RAR is compatible with this framework and non-diagnostic of its coefficient; claims in either direction are interpolation-shape and mass-to-light artefacts, and two such claims in the author’s own corpus were withdrawn for exactly this reason.

The same caution applies to the BTFR intercept, which is Υ -degenerate. The sharpest Υ -free handle — the gas-dominated slope — gives a box $[0.84, 1.36] \times 10^{-10}$ that contains both 9.36×10^{-11} and 1.2×10^{-10} and does not discriminate.

5.2 Lensing: this is what the switch buys

The metric is modified, so light feels the enhancement. The 21σ exclusion of §0 does not apply. **This is the entire reason for the paper**. The lensing *amplitude* is a prediction of the relativistic completion, not of the non-relativistic theory written here — see §7.

5.3 Clusters: unchanged, and still the sharpest problem

The switch buys nothing here, and that must be said. The cluster shortfall is a property of the *kernel*, not of which sector is modified. On this framework’s own kernel the required boost at R_{500} has median $\eta = 2.334$ (0.368 dex), at 2.0–4.1 σ .

A full audit of the measurements [9] finds they largely hold up. There *is* one real, published, unresolved error in the X-ray mass scale — Chandra hydrostatic masses are $14 \pm 2\%$ higher than XMM’s on 64 clusters observed by both — and it points the framework’s way. **It is 14% against a needed 133%.** The better-established systematic (hydrostatic bias from uncounted non-thermal pressure) points the *other* way, and the two substantially cancel.

The wall is the cosmic baryon budget, and no instrument can move it. Clusters are already at the cosmic baryon fraction: $f_{\text{gas}}(r_{500}) = 0.163 \pm 0.032$ against $\Omega_b/\Omega_m = 0.167 \pm 0.006$. Headroom for undetected baryons is at most 1.22 $\times$ against a required 2.33 $\times$. Every knob turned favourably at once absorbs 1.39 $\times$, leaving a residual factor 1.68.

If you are looking for the observation most likely to kill this framework, it is clusters, and it has been for thirty years.

5.4 Wide binaries: the registered prediction moves

The author maintains a frozen, hash-stamped Gaia DR4 pre-registration. **Its target is the modified-inertia value, and this arm predicts a different one.** For a point-field AQUAL-type external-field effect the boost is isotropic and equals $\sqrt{\nu}$ at the external field:

$$\gamma_v^{\text{MG}} = \sqrt{\nu(y_{\text{ext}})} = 1.2139 \text{ (canonical), } 1.2592 \text{ (alt footing)}$$

against $\gamma_v^{\text{MI}} = 1.1582$, range 1.1311–1.1964. **The ranges are disjoint.** Moving the registered target is an amendment-grade change and has not been made in this paper.

It cuts both ways: the separation is 2.68 σ at the registered sample size, so **DR4 can distinguish the two arms** — which the modified-inertia reading could not do. That is a lever, not a decisive test, and it is stated as one.

5.5 The directional external-field test: now a signal, and it already fired

This is the strongest thing the switch buys, and it required no new observation. In an aligned rotation-curve asymmetry relative to the external field direction, **pure modified inertia predicts exactly zero**; AQUAL-class theories predict 1–4% with a definite sign. A first firing of that test gave $\hat{A} = +2.95$ at $p = 0.029$ **with the AQUAL-class sign.**

So a measurement that was evidence *against* the previous arm is evidence *for* this one. At $p = 0.029$ it is a lever rather than a confirmation; settling it needs $N \sim 1157$ galaxies, and the field direction is reconstructable from existing peculiar-velocity maps.

6. What the switch cost

Stated first-class, because two of these were the headline results of the arm being abandoned.

- $a_0 = \frac{2}{3}c m^2/g$ **is gone**. That reading was the modified-inertia memory kernel’s first moment. A Bekenstein–Milgrom theory has no kernel and no moment; a_0 enters $\mathcal{F}$ directly.
 - **And with it the ζ -pole no-go**, which was a theorem about that first moment. Its conclusion — that a_0 has the status of a renormalisation condition — may still hold here, but it would have to be **re-derived** and is not inherited.
 - **The g^{-2} Lorentz-violation prediction is gone in pure Bekenstein–Milgrom**, which has no preferred frame at all. It survives only if the relativistic completion is AeST-type, carrying a vector. **Which completion is adopted therefore decides whether the framework keeps that prediction — it is not a detail.**
 - **The Cassini quadrupole tension is inherited**. The AeST-class realisation carries a 3–15 σ RAR-versus- Q_2 tension that modified inertia did not. **This is the item a referee will find first.**
 - The **parity theorem** — that no polynomial-in- u worldline action can produce MOND — remains *true* but is no longer load-bearing.
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7. What is owed

1. **A relativistic completion**. The theory as written is non-relativistic. Cosmology, the CMB, and the lensing *amplitude* all require AeST or TeVeS [10,11], each with its own constraints. **Until that is fixed, “it lenses” is a statement about the sign and the mechanism, not about the amplitude.**
 2. **The amendment moving γ_v** (§5.4), which is a registration change and deliberately not made here.
 3. **Clusters** (§5.3). Unchanged by the switch and still the most likely killer.
 4. $\kappa = \frac{1}{2}$. Sixteen distinct attempts to force it have failed; the residue after tracing $\sqrt{6}$ to the Friedmann equation and $\sqrt{\pi}$ to the three-dimensional momentum measure is a single bare factor of two.
 5. **Whether the g^{-2} prediction survives**, which follows from item 1.
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8. What is not claimed

- **Bekenstein–Milgrom is not new**. It is theirs, 1984. The interpolating function is Milgrom’s, 1999. What is offered here is the coefficient, the kernel forced by ranging, and the closed-form free function with its health proved for that kernel.
 - $\kappa = \frac{1}{2}$ **is fitted, not derived**.
 - **No claim is made regarding particle physics, the Standard Model, or unification**. The author publicly withdrew earlier statements of that kind in June 2026 and does not restate them. The relevant obstruction is specific: this framework’s characteristic constant carries $\sqrt{\pi}$ while all flavour observables are algebraic, so the structure is *blind* to the gauge sector.
 - **This is modified gravity, and it does not derive a_0** . The accompanying script says so in its own exit message.
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9. Reproducibility

Every quantitative claim is produced by a committed script that exits non-zero if any internal check fails, and each carries negative controls that must trip.

- `mi_route_a_field_theory_2026.py` — §§3–4, 11/11 checks.
- `mi_route_a_kernel.py` — §2, 6/6 checks.
- `mi_mg_arm_standing_2026.py` — §§5.4–6, 18/18 checks.
- `mi_lensing_axis_2026.py` — §0, 24/24 checks.
- `mi_finsler_null_cone_2026.py` — §0, 23/23 checks.
- `mi_cluster_measurement_audit_2026.py` — §5.3, 27/27 checks.
- `mi_route_a_wb_gamma_v_2026.py` — §5.4, 26/26 checks.

Both a_0 footings (canonical ρ_{DE} with cH_Λ ; alternative $\times 1.2048$) are carried on every dimensionful number.

AI-assistance disclosure. Portions of the analysis, numerical verification and drafting were carried out with the assistance of a large language model (Anthropic Claude). The author directed the work, specified and reviewed every load-bearing calculation, and takes full responsibility including for any errors. No AI system satisfies the criteria for authorship and none is listed as an author. Claims found to be incorrect during this work were withdrawn rather than quietly re-scoped; the modified-inertia arm’s demotion, recorded in §0 and §6, is the largest of them.

References

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